

# **Cognitive AI Project – Conversational Upgrade Notes**

## **Current System**

User enters one problem. ML prediction runs. Emotion and sentiment detection run. Simulation and planner run. Phi-2 generates one final mentor-style response.

## **Target Conversational System**

Multi-turn chat like ChatGPT. AI remembers previous messages. AI asks follow-up questions. User can request custom plans like 2-week or 3-week plans. More human and personalized counseling experience.

## **Main Files Required**

ui/app.py for chat-style Streamlit UI using st.chat\_input and st.chat\_message.  
llm/generate\_response.py for context-aware Phi-2 response generation using chat history.  
simulation/planner.py for dynamic planner supporting 2-week, 3-week and 4-week plans. Optional  
memory.py for tracking financial pressure, goals and emotional state.

## **Example Conversation Flow**

User says they want CAT preparation but family wants a job. AI asks about financial urgency. User says urgency is high. AI suggests short-term job plus CAT preparation. User asks for a 3-week plan. AI generates a personalized 3-week action plan.

## **Final Recommendation**

Keep training on Colab. Use Phi-2 with LoRA adapter. Run inference locally using RTX 3050 GPU with Python 3.10 CUDA setup. Use Streamlit for final demo. Convert the project into a full conversational AI system for stronger MCA presentation.